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Project Title: Diagnosing Water User Response to the 2026 Drought and Improving Preparedness Via Tailored Drought Reports

Project Type: Drought Working Group

Geographic Focus of Study: Statewide

Idaho Research Priority Topic: Hazards & Climate – Drought (Statewide), with connections to Technology & Innovation – Water Monitoring & Data Accessibility.

Technical Details

Project Description

The 2026 snow drought left much of Idaho with severely reduced streamflow, forcing reduced surface-water diversions by irrigation and canal companies. Groundwater users face a related constraint since many of them operate under a mitigation plan (2024 Stipulated Mitigation Plan) that requires compliance with groundwater conservation obligations set at the groundwater district scale within rolling four-year compliance blocks. The 2026 water supply did not support historical cropping intensity under those obligations, at least based on anecdotal evidence. Farmer testimonies show that they adapted to these conditions heavily, e.g., by fallowing land, switching to lower water-use crops, and green-chopping small grains rather than carrying them to full maturity. However, the magnitude, spatial pattern, and consumptive use consequences of these changes have not been systematically documented. At the same time, the water users navigating these constraints have no single, spatially relevant source of real-time drought information: existing products are scattered across agencies and reported at county or climate-division scales that do not match the irrigation- and groundwater-district boundaries that actually govern their water decisions.

This project combines a (1) retrospective diagnosis of the 2026 irrigation season with a (2) prospective, operational drought-reporting tool, giving Idaho agencies and water managers both an accounting of what happened this year and a durable capability for tracking what happens next.

Objective 1 (Retrospective): Postmortem diagnosis of water user response to 2026 Irrigation Season

How did cropping systems and field-scale consumptive water use in Idaho's irrigated and dryland agriculture change during the 2026 snow drought relative to historical patterns, and what do those changes reveal about how producers adapted to reduced water supply and mitigation-plan obligations? Using field-scale OpenET evapotranspiration data and USDA Cropland Data Layer (CDL) crop-type classification at 30 m resolution combined with IDWR field polygons, we will quantify how consumptive water use in Idaho's irrigated fields in 2026 departed from historical, multi-year ET baselines at the field, crop-type, and district (surface- and groundwater-district) scale. We hypothesize that reduced soil moisture, surface-water diversions, groundwater conservation obligations, possible curtailments and farmer-driven

fallowing and cropping changes produced a measurable, mappable reduction in growing-season ET relative to normal years. Analysis will proceed incrementally as OpenET data release monthly through the remainder of 2026, with the crop-type-resolved analysis finalized once the 2026 CDL product is released (typically the following February).

Objective 2 (Prospective): Real-Time, water user-tailored drought reporting

We will build a public webpage delivering automated drought reports, updated every five days tailored to Idaho surface-water entities and groundwater districts. Reports will be generated on the Climate Engine's codebase, reconfigured around our own district boundary layers rather than default administrative boundaries. We are also open to exploring other meaningful spatial units for Idaho stakeholders (e.g. tribal lands; water districts etc.). Each report will synthesize precipitation-based drought indices, evaporative demand-based drought/anomaly indices, and U.S. Drought Monitor category status into consistent graphics, accompanied by plain-language guidance explaining what each report shows and how to interpret it.

Description of Applied Outcomes

Intended users and decisions: Objective 1 outcomes help IDWR, districts, and the legislature understand how agriculture and water users respond under extreme shortage, a stress test revealing lessons for future drought years. Objective 2 outcomes help individual water users and district managers to gain tailored, jurisdiction-specific drought early warnings instead of county-level data that mixes in large non-irrigated/desert areas.

When used in the decision cycle: Objective 1 is timed to inform FY2027 legislative reporting and pre-season planning of the 2027 irrigation season. Objective 2 outputs are used continuously in-season for near-real-time awareness and preparedness, most critically during the April–September irrigation season.

Readiness level at project end: Objective 1 will reach a completed, publication-ready synthesis (manuscript submission plus contribution to the statewide white paper). Objective 2 will reach a fully deployed public webpage (before Spring 2027) operating on a five-day cadence for the water user entities.

Methodological Approach

- **Retrospective (Objective 1):** Field-scale OpenET time series over IDWR field polygons, joined to CDL crop type, compared against multi-year historical ET baselines using anomaly detection by crop type, field, and district. Most analysis will be conducted in Google Earth Engine and findings will be compiled as an Earth Engine application.
- **Prospective (Objective 2):** We will customize Climate Engine template built on IDWR-sourced irrigation- and groundwater-district boundaries. Each Climate Engine drought report packages, for a given area of interest, a short-term drought blend (PDSI, Palmer Z-Index, SPI-30d, SPI-90d; ~30 days–9 months) and a long-term drought blend (PDSI, Z-Index, SPI at 6mo/1yr/2yr/5yr; 6 months–5 years) rendered as a color-coded current-conditions map, a summary table of drought-class area/percent now vs. 3 months ago vs. 1 year ago, and a 1985–present time-series chart for historical context. We will host these graphics and information on a webpage, which will be handed over to UI RCDS for further front-end development.

Proposed Deliverables

- **Peer-reviewed publication:** Objective 1 is a spatial subset of a larger-scale manuscript planned on agricultural producers' response to 2026 drought across the West.
- **Public-facing products:** Water users drought reports tool updated at a 5 day cadence (objective 2). For objective 1, we will produce an Earth Engine app allowing users to interface with spatial data on grower response and impacts on CU.
- **Statewide synthesis contribution:** The findings on 2026 cropping adaptation and consumptive-use impacts will be contributed to the Drought Working Group's statewide synthesis white paper.
- **Coordination with stakeholders:** We will rely on IWRRI to facilitate coordination meetings to gather feedback and steer the projects. Stakeholders will likely vary for the two objectives, but we will plan for 2-3 meetings overall.
- **Presentations:** We will rely on IWRRI to let us know what the best groups would be to present to. Some ideas are ID Water Users Association; Fall Water Supply Meeting; Water Quality Workshop
- **Data and code** deposited to HydroShare by project end.

Timeline (1 Year)

Months 1-3:

- Water user drought reports tool prototype ready.
- 1 stakeholder meeting to discuss initial plans and roadmaps to deliver products.
- Develop ET and crop type climatologies for anomaly detection for 2026 growing season

Months 4-6

- November-December 2026: Complete record of 2026 irrigation season OpenET data becomes available; develop anomalies at the field scale and evaluate divergence from historical climatologies.
- Act on initial feedback on water user drought reports tool

Months 6-9:

- February 2027: USDA Cropland Data Layer 2025 becomes available allowing for crop adaptation analysis and tying it back to ET analysis.
- Deliver a finalized version of water user drought reports tool after collaboration with RCDS.

Months 9-12: Primarily reporting; stakeholder communication; presentations; and submitting manuscript.

Future Development Opportunities

Being in the drought working group, I'd like to engage with NOAA NIDIS staff to understand regional priorities in the area of evaporative demand-based drought indicators (e.g. thirstwaves). I will be interested in exploring NOAA NIDIS and Climate Program Office grants. We are currently also working on a recently funded USDA NIFA Rapid Response grant in a similar space and there will be ways to co-produce some outcomes.